

PHOENIX SKY HARBOR, AZ, USA

WMO: 722780

Lat: 33.428N

Lon: 112.004W

Elev: 337

StdP: 97.34

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23183

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	4.0	5.5	-16.5	0.9	19.0	-14.0	1.2	19.2	8.8	13.5	7.5	15.0	1.6	100	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	43.6	20.7	42.5	20.6	41.5	20.5	24.2	34.2	23.8	34.3	23.3	34.5	4.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.3	27.7	20.9	16.2	28.9	19.8	15.1	30.2	74.5	33.8	72.7	34.2	70.9	34.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.4	6.4	DB	1.4	46.0	1.5	1.3	0.3	47.0	-0.6	47.7	-1.4	48.5	-2.5	49.4	(4)
(5)			WB	-1.9	25.2	1.5	1.0	-3.0	25.9	-3.8	26.5	-4.6	27.1	-5.7	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.2	13.8	15.4	19.1	22.6	27.5	33.0	35.1	34.5	31.6	24.9	18.4	13.3
	DBStd	8.49	3.02	3.32	3.67	3.86	3.64	2.99	2.45	2.43	2.81	3.76	3.71	2.99
	HDD10.0	15	6	2	0	0	0	0	0	0	0	0	1	7
	HDD18.3	486	142	92	36	11	1	0	0	0	0	3	44	158
	CDD10.0	5182	124	153	283	379	543	691	779	758	648	463	253	108
	CDD18.3	2610	2	9	60	139	285	441	521	500	398	208	47	1
	CDH23.3	40023	29	119	688	1661	4031	7407	8904	8359	6038	2346	427	14
CDH26.7	25997	1	16	226	751	2352	5159	6447	5909	3860	1157	118	1	
Wind	WSAvg	2.8	2.1	2.5	2.8	3.3	3.2	3.2	3.4	3.1	2.8	2.5	2.2	2.1
Precipitation	PrecAvg	188	20	19	22	6	3	2	22	24	17	14	16	22
	PrecMax	387	133	80	80	34	30	43	131	90	85	112	71	101
	PrecMin	72	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	74	22	20	20	8	7	6	23	22	20	21	17	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	28.8	33.7	36.6	41.3	45.3	45.4	44.5	42.2	37.8	32.0	25.0
		MCWB	11.6	12.6	14.8	15.8	18.3	20.1	21.1	21.3	20.7	18.5	14.7	11.8
	2%	DB	23.9	26.6	30.9	34.7	39.2	43.1	43.8	43.1	40.8	35.9	29.7	23.1
		MCWB	10.9	12.2	14.0	15.1	17.3	19.0	21.1	21.5	20.3	17.7	14.2	11.5
	5%	DB	22.2	24.7	29.2	32.9	37.5	41.9	42.6	41.8	39.5	34.3	27.8	21.3
		MCWB	10.4	11.6	13.3	14.4	16.6	18.8	21.2	21.6	20.1	17.1	13.7	10.7
	10%	DB	20.4	22.7	27.4	31.0	35.9	40.7	41.4	40.6	38.2	32.6	25.9	19.5
		MCWB	9.9	11.1	12.7	13.8	16.1	18.5	21.1	21.7	20.2	16.4	13.1	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.1	15.8	17.2	19.6	22.8	24.9	25.1	24.4	22.1	17.2	14.9
		MCDB	18.7	22.1	28.5	33.2	36.2	37.7	34.1	35.7	32.9	29.5	25.4	19.6
	2%	WB	13.1	14.0	15.1	16.1	18.7	21.8	24.1	24.3	23.5	20.1	16.0	13.5
		MCDB	18.4	20.9	27.5	32.0	35.3	37.1	34.1	34.7	33.1	30.5	24.3	19.0
	5%	WB	12.1	13.1	14.2	15.2	17.8	21.0	23.7	23.9	22.9	19.0	15.0	12.4
		MCDB	18.4	20.7	25.9	30.2	34.6	36.7	34.3	34.7	33.2	30.2	24.2	18.0
	10%	WB	11.2	12.2	13.4	14.4	16.8	20.0	23.2	23.5	22.3	17.9	14.0	11.3
		MCDB	17.9	20.2	24.9	29.3	33.7	37.2	34.5	34.8	33.0	29.6	23.9	17.4
Mean Daily Temperature Range	5% DB	MDBR	11.5	11.5	12.6	12.9	13.1	13.5	11.4	11.1	11.6	12.4	12.0	11.0
		MCDBR	14.1	14.6	15.3	15.2	14.9	14.5	12.3	12.1	12.7	14.0	14.3	13.3
		MCWBR	6.0	6.0	5.3	5.3	4.8	4.2	3.3	3.1	3.7	4.9	5.5	5.7
	5% WB	MCDBR	10.3	11.7	13.7	13.9	13.7	12.0	10.8	10.9	10.1	11.0	11.8	10.0
		MCWBR	4.9	5.1	5.0	5.2	4.8	4.0	2.8	3.0	3.3	4.5	5.2	4.8
Clear-Sky Solar Irradiance	taub	0.284	0.290	0.317	0.326	0.335	0.346	0.408	0.394	0.350	0.325	0.294	0.284	
	taud	2.555	2.547	2.450	2.413	2.419	2.388	2.346	2.373	2.467	2.487	2.531	2.559	
	Ebn at Noon	943	971	961	961	949	933	874	883	914	915	922	923	
	Edh at Noon	80	90	107	116	117	120	125	120	104	94	81	76	
All-Sky Solar Radiation	RadAvg	3.39	4.34	5.84	7.27	8.15	8.44	7.40	6.82	6.11	4.92	3.76	3.07	
	RadStd	0.25	0.34	0.28	0.20	0.27	0.31	0.33	0.22	0.23	0.28	0.16	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.53	N/A	N/A	+0.26	N/A	N/A	N/A	-59	+201	+140
(47)	Regional (13 neighbors)	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	-81	+98	+30

Nomenclature: See separate page